

## Smart Lender – Project Flow

### Overview

The **Smart Lender – Loan Approval Prediction System** is developed using a structured workflow divided into **five Epics**. Each Epic focuses on a critical stage of the project, including data preparation, machine learning model development, system integration, testing, and deployment. This approach ensures a systematic, modular, and efficient development process while maintaining high prediction accuracy and system reliability.

---

### Epic 1: Data Collection and Requirement Analysis

#### Objective

Collect and analyze the dataset required for loan approval prediction.

#### Activities

- Gather historical loan application data.
- Identify important features such as applicant income, education, employment status, loan amount, loan term, and credit history.
- Define business requirements and prediction objectives.
- Analyze data quality and identify missing or inconsistent values.

#### Deliverables

- Loan dataset
  - Requirement specification
  - Feature list
- 

### Epic 2: Data Preprocessing and Feature Engineering

#### Objective

Prepare the dataset for machine learning.

#### Activities

- Remove duplicate records.

- Handle missing values.
- Convert categorical data into numerical format.
- Normalize or standardize numerical attributes.
- Select the most relevant features for prediction.
- Split the dataset into training and testing datasets.

### **Deliverables**

- Clean dataset
  - Processed features
  - Training and testing datasets
- 

## **Epic 3: Machine Learning Model Development**

### **Objective**

Build and train the loan approval prediction model.

### **Activities**

- Select appropriate machine learning algorithms.
- Train the model using historical loan data.
- Evaluate multiple algorithms.
- Measure model performance using accuracy, precision, recall, and F1-score.
- Select the best-performing model.

### **Deliverables**

- Trained machine learning model
  - Model evaluation report
  - Performance metrics
- 

## **Epic 4: System Integration and Prediction**

### **Objective**

Integrate the trained model with the Smart Lender application.

### **Activities**

- Connect the machine learning model with the application.
- Accept applicant information through the user interface.
- Process user input.
- Generate loan approval predictions.
- Display prediction results with confidence scores.

### **Deliverables**

- Integrated prediction system
  - User interface
  - Prediction output
- 

## **Epic 5: Testing, Validation, and Deployment**

### **Objective**

Validate the system and prepare it for deployment.

### **Activities**

- Perform functional testing.
- Validate prediction accuracy using test data.
- Fix identified issues.
- Optimize system performance.
- Deploy the Smart Lender application.

### **Deliverables**

- Tested application
- Validation report
- Final deployed system